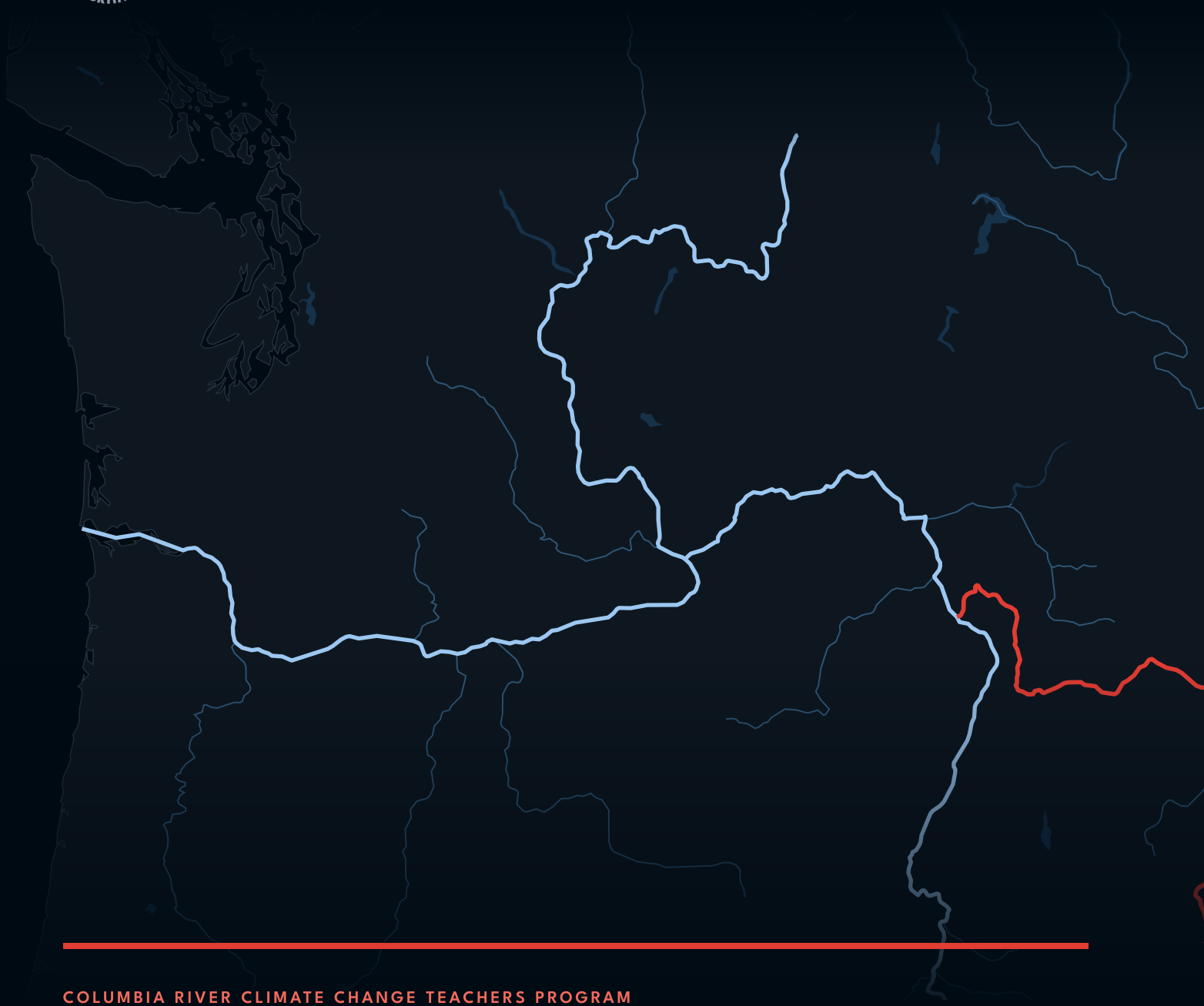


COLUMBIA RIVER CLIMATE CHANGE TEACHERS PROGRAM

UNDERSTANDING A CLIMATE OF CHANGE

GRADES 6 TO 12 HANDS-ON LESSONS

Patrick Freeland and Cayla Jones · ATNI Climate, Affiliated Tribes of Northwest Indians

Workshop printing, 10/20/2026 · Draft for review

FOUR QUESTIONS, ONE RIVER

1	2	3	4
How does a cloud work?	How do you know what you know?	What can we do?	How do we work together?

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Columbia River Climate Change Teachers Program | Affiliated Tribes of Northwest Indians (ATNI) Climate

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Built on *Understanding a Climate of Change: Understanding an Ever Changing Climate*, by Patrick Freeland and Cayla Jones, and on Steve Robinson's *Climate Change and the Tribes: Columbia River Edition* (draft).

PLACEHOLDER: PF opening in his voice

Printed 10/20/2026 from the Markdown source in the public repository https://github.com/atniclimate/climate-curriculum. Live workshop: https://atniclimate.github.io/climate-curriculum/

How to Use This Document

These lessons are for teachers who have completed the 30-minute workshop built on the four parts of *Understanding a Climate of Change*: (1) How does a cloud work? (2) How do you know what you know? (3) What can we do? Adaptation, Mitigation, Resilience; and (4) How do we work together? Relationships Matter. The four units follow the same parts, so you can walk from the workshop into your classroom with one structure. An opening and a closing lesson hold them together, the way a circle holds a conversation.

This set carries forward the Grades 6 to 12 tier of the Climate Change Education Pathway (the Changing Currents Classroom Edition). It keeps what worked at the 2025 Changing Currents Tribal Youth Water Summit (sketchbooks, the Elwha story, the Elder-and-child pitch, the career gallery, and the closing circle) and replaces the field trip with a virtual field study and classroom models. Everything is set on the Columbia River, with the Elwha as a comparison story of restoration.

- **Shared spine:** Begin with experience, not abstraction; honor plural ways of knowing; use the Elder-and-child test; treat data literacy as sovereignty; move from learning to leading; relationships matter.
- **Sequence and time:** 14 lessons, most one period of 45 to 55 minutes; three are two-period (17 periods total). Short path: Opening, 1.1, 2.1, 3.1 Part A, 3.2, 4.1, Closing.
- **Sketchbooks:** One per student for the whole set; it is the portfolio. Sketching slows students down and lets everyone take notes, whatever their reading level. Do not grade the art.
- **Reflection, not ratings:** Every lesson ends with three questions: What did you learn? What idea shifted? What one question will you carry forward? Read answers for content; there is no score.
- **Placeholders:** Anything marked `PLACEHOLDER: ...` must come from Patrick, the teacher, or a Tribe. Anything marked "verify" was current when drafted and may have changed.

Climate Maturity, Hope, and Action

The NESC 215 Climate Change in Indian Country course at Northwest Indian College (NWIC) names a practice called climate maturity: taking care of yourself when dealing with future scenarios. Students need it too, especially when the material touches their own families, rivers, and foods. Steve Robinson's *Climate Change and the Tribes: Columbia River Edition* draft names its purpose in two words: hope and action. The workshop puts it this way: "The impacts are already here, so we respond quickly and on purpose, without getting lost in fear, the future, and the 'black mirror' of media."

- **Start with experience:** Hands first, then ideas.
- **Name feelings briefly:** "One word for how you feel right now" is enough. Feelings are information.
- **End on action or relationship:** No lesson ends on a projection.
- **Pass is always okay.**
- **Know your supports:** For some students, dams, fires, floods, and broken promises are family history. Know your counselor.
- **Bring numbers home:** Scale global figures to the river, the school, and the next step.

Unit Map

LESSON	TITLE	HANDS-ON CORE	GRADES	PERIODS	WORKSHOP PART (SHORT NAME)
Opening	Where We Come From	Talking circle; half-circle timeline (past)	Both	1	All

LESSON	TITLE	HANDS-ON CORE	GRADES	PERIODS	WORKSHOP PART (SHORT NAME)
1.1	Cloud in a Bottle and the Dripping Air Conditioner	Pressure cloud; condensation cups; dew point can	Both	1	1. Cloud
1.2	Watching the Atmosphere Move	earth.nullschool live; weather journal	Both	1	1. Cloud
1.3	Snow Is a Savings Account	Ice tray versus rain tray runoff model	Both	1	1. Cloud
2.1	Source Check	Climate claim card sort	Both	1	2. Knowing
2.2	Knowledge Across Generations	Permission-based interview plan and practice	Both	1 plus take-home	2. Knowing
2.3	The Elder and the Child Test	Story of Self, Us, and Now 60-second pitch	Both	1	2. Knowing
3.1	The Salmon Journey	A: temperature and oxygen demo; B: migration relay	Both	2	3. Adaptation
3.2	Adapt, Mitigate, or Build Resilience?	Venn sort of real Columbia Basin projects	Both	1	3. Adaptation
3.3	Virtual Field Study	Dynamic Drought Module data stations; framing worksheet	Both	1	3. Adaptation
4.1	Who Is Connected to Whom?	Yarn relationship web	Both	1	4. Relationships
4.2	Two Rivers	Elwha film sketch-notes; comparison	Both	1	4. Relationships
4.3	Good Energy, Wrong Place?	Mock consultation on Goldendale and Pushpum	9 to 12 (6 to 8 short version)	2	4. Relationships
Closing	From Learning to Leading	Career gallery; commitments; half-circle timeline (future); closing circle	Both	2	All

Working With Tribes Respectfully

Read this before you teach.

- **Start with your local Tribe:** Contact the education, cultural resources, or natural resources department of the Tribe or Tribes nearest your school. Ask what they want taught, what they do not, and whether someone can visit. PLACEHOLDER: local Tribe(s) and department contact
- **Permission from some is not permission from all:** As the *Climate Change and the Tribes* draft (Robinson) says of the Talking Stick, "Permission from some Tribal leaders does not mean permission from all Tribal leaders."
- **Do not invent or borrow cultural content:** These lessons contain no Tribal stories, songs, ceremonies, or teachings, on purpose. Where a lesson needs them, a placeholder waits for a knowledge holder the Tribe recognizes.
- **Compensate knowledge holders:** Offer an honorarium, cover travel, and share back what students made.
- **Never ask a Native student to speak for a Nation.** A student may choose to share.
- **Treat knowledge as sovereign:** Use the CARE Principles for Indigenous Data Governance (Collective benefit, Authority to control, Responsibility, Ethics) and the OCAP principles (Ownership, Control, Access, and Possession).

Local Contexts Traditional Knowledge (TK) and Biocultural (BC) Labels show how communities mark permissions.

- **Handle maps with care:** Do not present Treaty or ceded-land boundaries as definitive jurisdiction without the Nation's guidance.
- **Use the right words:** Tribes are sovereign governments and rights holders. Tribal community members are relatives, community members, Tribal citizens, rights holders, or partners, never "stakeholders."

Teacher Safety Note for Hands-On and Outdoor Work

- **Flame and alcohol:** Matches and rubbing alcohol (Lesson 1.1) are teacher-only. Never use both together.
- **Water:** Warm tap water only. Keep trays away from electronics; mop spills at once.
- **Games:** Clear space for the relay; walk in tight rooms; watch for yarn at ankle height.
- **Outdoors:** Check the Air Quality Index (AQI) at AirNow.gov during fire season and alerts at weather.gov. Bring water and shade.
- **Rivers:** No student enters a river without trained staff and a site plan. Cold-water shock is a risk even in summer.
- **Tribal lands:** Visiting Tribal lands, fishing sites, or cultural sites requires the Tribe's permission.
- **Media:** Follow school media-release policy. Record interviews only with the speaker's clear permission.
- **Emotional safety:** Give a heads-up before heavy content, offer a pass, and know your counselor's schedule.

Opening Lesson

OPENING

Where We Come From and the First Half of the Circle

- **Grade band and time:** Both; 50 minutes
- **Key lesson:** Weather is what we live through day to day; climate is the pattern those days make over many years. Our memories are already a record.
- **Key takeaway:** Everyone in the room is a witness to weather and water. Learning starts with what we have experienced.

Materials: Sketchbooks; colored pencils; a talking piece; chart paper; a Columbia River Basin map.

Teacher prep: Arrange a circle. The *Climate Change and the Tribes* draft (Robinson) suggests a Talking Stick, with permission from several Tribal leaders for that curriculum, and cautions that permission from some is not permission from all. Ask your local Tribal liaison first; meanwhile use a neutral talking piece, such as a river stone. If a Tribal guest opens the circle, follow their lead.

Procedure:

1. **Welcome (0 to 5 min):** PLACEHOLDER: land and water acknowledgment developed with the local Tribe; no generic script. Norms: one voice, listen to understand, pass is okay, personal stories stay in the circle.
2. **The talking piece (5 to 10 min):** Whoever holds it speaks; everyone else listens fully. As the draft says, we have two ears and one mouth for a reason.
3. **Circle round (10 to 22 min):** Name, homeplace, and one connection to water (a river, a fishing spot, a flood, a swimming hole). Go first. Borrow the Youth Summit's message: you matter, and you are not alone.
4. **Weather and climate (22 to 27 min):** Weather is short-term conditions in the atmosphere. Climate is weather averaged over a long period (climate normals use 30 years).
5. **First half of the circle (27 to 42 min):** Students draw a large half circle across the top half of a page, from birth at the left end, arching over to today at the right. Along it they sketch five to eight memories tied to weather, water, or seasons: a snow day, a smoke day, the hottest day, a flood, a day on the river, a power outage. The bottom half stays blank until the Closing.
6. **Pair share (42 to 47 min):** Partners trade one memory; two or three volunteers share.
7. **Reflection (47 to 50 min)**

Discussion prompts: Have you noticed changes in weather, seasons, or the environment where you live? Which of your memories would an older relative recognize, and which might be new?

Differentiation:

- **Grades 6 to 8:** Post a list of memory prompts; three memories is enough.
- **Grades 9 to 12:** Label each memory "weather" or "possible pattern," and for one, name the data that could confirm it (the date plus the nearest station's record).

Formative check: One memory labeled as weather, with one sentence on what evidence would show whether it is part of a pattern.

Reflection: (1) What did you learn about your own experience with weather and water? (2) What idea about weather or climate shifted for you? (3) What one question about your homeplace will you carry forward?

1

UNIT 1

How Does a Cloud Work?

Key lesson (from the workshop): Weather is based on fundamental physics, here and throughout the universe. Pressure and water are always moving, never fixed or stuck at one height.

Key takeaway (from the workshop): The same physics, and the same gauges and data points that predict tomorrow's weather, also show us patterns and change over many years. With enough data, we look ahead on momentum, not speculation.

LESSON 1.1

Cloud in a Bottle and the Dripping Air Conditioner

- **Grade band and time:** Both; 50 minutes
- **Key lesson:** Clouds form when air cools enough that water vapor condenses onto tiny particles. Pressure and temperature move together.
- **Key takeaway:** A cloud is physics you can make on a desk, and the same physics drives rain on the Cascades and the drip from an air conditioner.

Materials: Per group of 3 to 4: a clear plastic bottle (1 or 2 liter) with cap; warm tap water; a cup of ice water and a cup of room-temperature water; paper towels; dark paper and a flashlight. Teacher only: matches or about a teaspoon of rubbing alcohol (never both); a shiny metal can; ice; a thermometer.

Teacher prep: Run the bottle demo once. The cloud is faint; a flashlight beam against dark paper makes it visible. Look up today's dew point for your town on weather.gov.

Procedure:

1. **Sky sketch (0 to 5 min):** Sketch one cloud. Ask: What is a cloud made of? Record ideas without correcting.
2. **Set the cups (5 to 10 min):** Each table gets an ice-water cup and a room-temperature cup, outsides dried. Predict, then leave them alone.
3. **Cloud in a bottle (10 to 25 min):** Pour an inch of warm water into the bottle, swirl, cap, squeeze, release: little happens. The teacher adds smoke from a just-blown-out match, or a few drops of alcohol. Squeeze hard, release fast: a cloud appears, then vanishes on the next squeeze. Sketch both states.
4. **Back to the cups (25 to 32 min):** Water beads on the ice cup; it came out of the air. This is the dripping air conditioner: warm, moist air crosses cold coils, cools below its dew point, and water condenses and drips. A cloud does the same when rising air expands and cools.
5. **Dew point can (32 to 40 min):** Add ice bit by bit to water in the metal can, stirring with the thermometer. The temperature when fog first forms outside is close to the dew point. Compare with weather.gov.
6. **Bring it home (40 to 46 min):** On the map, trace Pacific air pushed up the Cascades, cooling and dropping rain and snow, then sinking and drying on the east side. That rain shadow is why Yakima, The Dalles, and Pendleton are so dry.
7. **Reflection (46 to 50 min)**

Discussion prompts: Why did the cloud form on the release? Why did smoke matter? (Vapor needs something to condense on: dust, smoke, sea salt.) Where have you seen the dripping air conditioner in your life?

Differentiation:

- **Grades 6 to 8:** A four-panel sketch using evaporation, water vapor, condensation, and pressure.
- **Grades 9 to 12:** Explain the cloud through gas behavior (fast pressure drop, expansion, cooling). Add that a warmer atmosphere holds about 7 percent more water vapor per 1 °C of warming, one reason heavy downpours are getting heavier.

Formative check: A labeled four-panel sketch and one sentence on why the cloud formed on the release.

Reflection: (1) What did you learn more deeply about the weather? (2) What changed in how you think about clouds or the water cycle? (3) What one question about the weather do you want to answer?

LESSON 1.2

Watching the Atmosphere Move

- **Grade band and time:** Both; 50 minutes
- **Key lesson:** Pressure and water are always moving. Weather data collected over time becomes a climate record.
- **Key takeaway:** When you watch and record the sky carefully, you are doing what weather stations do, and your notes become part of a record.

Materials: Projector and internet; earth.nullschool.net; devices for pairs if available; sketchbooks; an outdoor thermometer; a rain gauge if you have one (a Community Collaborative Rain, Hail and Snow Network, or CoCoRaHS, gauge is ideal).

Teacher prep: On earth.nullschool.net, click "earth" to open the menu. Try Mode: Air, Height: Surface, with the mean sea level pressure (MSLP) overlay; then Height: 500 hPa; then the total precipitable water (TPW) overlay. Center on the Pacific Northwest. Find your local National Weather Service (NWS) forecast on weather.gov.

Procedure:

1. **Silent sky watch (0 to 5 min):** Three quiet minutes outdoors or at a window. Sketch the sky; note wind direction from flags or trees.
2. **Nullschool tour (5 to 20 min):** Find a low-pressure system spinning counterclockwise off the coast. Switch to 500 hPa for the fast flow aloft. Switch to TPW and look for a long, narrow band of moisture aimed at the coast (an atmospheric river). What happens when it reaches the Cascades?
3. **Pair missions (20 to 30 min):** Find the strongest wind near your town, the nearest low, and the 500 hPa wind above you. Sketch each.
4. **Launch the journal (30 to 40 min):** Make today's entry. Commit to ten entries over three to four weeks.
5. **Forecast check (40 to 46 min):** Read one line of the NWS forecast; predict tomorrow's entry.
6. **Reflection (46 to 50 min)**

Student handout: Weather Journal Page

DATE AND TIME	SKY SKETCH	CLOUDS	TEMPERATURE	WIND (DIRECTION, FEEL)	PRECIPITATION	WHAT NULLSCHOOL SHOWS	ONE QUESTION

Discussion prompts: Why does the atmosphere never sit still? If one day is weather, how many days before you can talk about climate?

Differentiation:

- **Grades 6 to 8:** A cloud picture chart (cumulus, stratus, cirrus, cumulonimbus); entries can be mostly sketches.
- **Grades 9 to 12:** Compare entries with the nearest official NWS or CoCoRaHS station and record differences. Discuss 30-year normals and why one warm week is not a trend.

Teacher note: Nullschool shows model data updated every few hours. It is for seeing, not warning; use weather.gov for warnings.

Formative check: A complete first entry and one sentence on a pattern seen on the live map.

Reflection: (1) What did you learn about how the atmosphere moves? (2) What shifted in how you see the sky over your homeplace? (3) What question will your journal help you answer?

LESSON 1.3

Snow Is a Savings Account

- **Grade band and time:** Both; 50 minutes
- **Key lesson:** In the Columbia Basin, snowpack stores winter water and releases it slowly into spring and summer. Warmer winters turn more snow to rain and melt snow earlier.
- **Key takeaway:** Same water, different timing, very different river. Timing matters for salmon, farms, and people.

Materials: Per group: two identical trays, each propped on a book to drain into a measuring cup; about two cups of ice; the same volume of water; a sponge (extension); a timer; graph paper.

Teacher prep: Make ice the day before. Melt one batch to learn how much water it holds, and give the rain tray that exact amount. Open the snowpack layer on the Dynamic Drought Module (DDM) or the Natural Resources Conservation Service (NRCS) Snow Telemetry (SNOTEL) map and pick a station near your school.

Procedure:

1. **Hook (0 to 7 min):** Show the SNOTEL station. Snow water equivalent is how much water is in the snow if you melted it.
2. **Set up (7 to 12 min):** Tray A is the rain year: pour water at the high end. Tray B is the snow year: pile ice at the high end. Both drain into cups labeled "river."
3. **Measure (12 to 37 min):** Record each cup every five minutes. Meanwhile, teach the water year: most precipitation falls October to March, and east of the Cascades snowmelt carries rivers through dry summers. In 2015, many places got near-normal precipitation, but it fell as rain: a snow drought, with low, warm rivers by summer. At 20 minutes, add a sponge at the bottom of Tray A for wetlands, beaver ponds, and floodplains that hold water back.
4. **Graph (37 to 45 min):** Plot both trays on one set of axes.
5. **Reflection (45 to 50 min)**

Student data sheet

TIME (MIN)	RAIN TRAY (ML)	SNOW TRAY (ML)	RAIN TRAY WITH SPONGE (ML)	NOTES
0, 5, 10, 15, 20, 25				

Discussion prompts: Which river still has water in August? What would a salmon, a farmer, a dam operator, and a family fishing at a usual and accustomed place each want this graph to look like?

Differentiation:

- **Grades 6 to 8:** Bar graph at three time points; sketch both rivers in August.
- **Grades 9 to 12:** Cumulative line graph and melt rate in mL per minute. Open a real U.S. Geological Survey (USGS) record for the Yakima or Umatilla River (DDM streamflow layer or USGS Water Data) and find the snowmelt peak.

Teacher note: Ice cubes melt far faster than a mountain snowpack. The model shows the pattern, not the timing.

Formative check: Two predicted hydrographs, a snow year and a warm rain year, on one set of axes with labeled peaks.

Reflection: (1) What did you learn about snow and rivers? (2) What idea about water shifted for you? (3) What one question about your nearest river will you carry forward?

2

UNIT 2

How Do You Know What You Know?

Key lesson (from the workshop): Climate is social as much as it is scientific; society moves action much more than the science does. How we come to know things, and who we trust, shapes what we believe.

Key takeaway (from the workshop): Narrative and emotion move us whether we notice it or not, so we learn to read both our sources and ourselves. Think about thinking.

LESSON 2.1

Source Check: Who Says It, and Why?

- **Grade band and time:** Both; 50 minutes
- **Key lesson:** Every claim has an author, a purpose, and a way of knowing, and every message carries emotion.
- **Key takeaway:** Read your sources and read yourself. Honest sources can still be wrong, and a strong feeling is a signal to check, not a reason to share.

Materials: One card set per group of 3 to 4; chart paper sorting mats with four columns; colored dots; devices (optional).

Teacher prep: Print and cut the cards without the teacher notes column.

Procedure:

1. **Quick write (0 to 5 min):** Name one thing you know about climate. How do you know it? Who told you?
2. **Model (5 to 12 min):** On one card, model the five Source Check questions: Who says it? How do they know? What do they want me to do or feel? What do I feel reading it? Can I check it?
3. **Sort (12 to 30 min):** Columns: "Checks out," "Honest but wrong or incomplete," "Pushing an agenda," "Need more information." Then dot every card that uses strong emotion.
4. **Reveal (30 to 40 min):** Walk through the notes. Linger on Card 1 (a unit error in a well-meaning handout) and Card 3 (emotion grounded in evidence). Emotion does not make a claim false.
5. **Think about thinking (40 to 46 min):** Each student writes which card they wanted to believe and why.
6. **Reflection (46 to 50 min)**

Claim cards and teacher notes

#	CLAIM	SOURCE AS PRESENTED	TEACHER NOTES
1	"Earth's average temperature has risen by as much as 33.8 °F since the Industrial Revolution."	A well-meaning classroom handout	Honest but wrong: 1 °C converted as a thermometer reading. The recent decade is about 1.1 to 1.3 °C (2.0 to 2.4 °F) above the 1850 to 1900 average.
2	"2024 was about 1.55 °C above the pre-industrial average."	World Meteorological Organization (WMO)	Checks out for that one year. One year is not the long-term average.
3	"The truth is that extinction of salmon populations is happening now."	Shannon Wheeler, Chairman, Nez Perce Tribe, statement of 06/12/2025 (compiled by the Columbia River Inter-Tribal Fish Commission, CRITFC)	Strong emotion and strong evidence together. What data would you look for?
4	"It's too late. Nothing we do matters now."	A viral post	Not supported: every fraction of a degree avoided lowers risk. Who benefits when people give up?
5	"The climate has always changed, so humans can't be causing it now."	A comment at a public meeting	Does not follow. The Intergovernmental Panel on Climate Change (IPCC) finds human influence on current warming unequivocal.
6	"Hydropower is clean, so the dams have no downside."	An energy ad (written for this lesson)	Incomplete. Low-carbon power, but dams block and warm rivers and flooded Celilo Falls.
7	"Salmon need water below 60 degrees to survive."	A classroom poster	Oversimplified. Stress rises as water warms; adult migration is impaired above roughly 20 to 21 °C (68 to 70 °F).

#	CLAIM	SOURCE AS PRESENTED	TEACHER NOTES
8	"In summer 2015, roughly 250,000 adult sockeye died in warm water in the Columbia Basin."	Fisheries agencies and CRITFC	Checks out. Find the original source.
9	"Our new natural gas plant is part of the clean energy future."	A press release (written for this lesson)	Agenda. Less carbon than coal, still emissions and methane leaks. Clean compared with what?
10	"The era of global warming has ended; the era of global boiling has arrived."	United Nations Secretary-General António Guterres, press conference, 07/27/2023	Rhetoric: real direction, not literal. Does it move you to act or to shut down?
11	"My grandmother says the berries ripen earlier than when she was young."	A student's family (written for this lesson)	Lifetime observation. It fits no column neatly; let students argue how to honor it.
12	"Tribes have contributed little to climate change but feel the impacts first."	The *Climate Change and the Tribes* draft (Robinson); Fifth National Climate Assessment, Chapter 16	Impacts are well documented. What data would measure contribution?

Discussion prompts: Can a source be honest and wrong at once, and what should it do when it finds the error? When a message makes you feel something strongly, what is your next move?

Differentiation:

- **Grades 6 to 8:** Cards 1 to 8, three columns (drop "Need more information").
- **Grades 9 to 12:** All 12. Practice lateral reading: look up who the source is before judging the claim. Rewrite Card 1 or 7 as one accurate sentence an Elder and a child would both understand.

Formative check: Read a new claim aloud; each student writes a one-line Source Check (who, how they know, what they want, how to check).

Reflection: (1) What did you learn about how we come to know things and whom we trust? (2) Was there an idea you held that changed or became clearer? (3) What one question about climate change do you want to follow up on?

LESSON 2.2

Knowledge Across Generations

- **Grade band and time:** Both; 50 minutes, plus a take-home interview due before Lesson 4.1
- **Key lesson:** Knowledge lives in people and places. It passes across generations and carries responsibilities about how it is shared.
- **Key takeaway:** Asking well is a skill. Permission, listening, and giving back come before recording anything.

Materials: Interview protocol (below); sketchbooks; a family letter explaining the assignment.

Teacher prep: If possible, invite a knowledge holder through the local Tribe's education or cultural department to model or advise, and compensate them. PLACEHOLDER: invited knowledge holder, arranged through the Tribe Send the family letter and check school policy on recordings. Students without access to an Elder may interview any older adult in their life or use an oral history the Tribe itself has published. PLACEHOLDER: Tribe-approved published oral histories

Procedure:

1. **Circle question (0 to 8 min):** Who taught you something about weather, water, or the land that was not in a book?
2. **Why permission matters (8 to 15 min):** Some knowledge is meant to be shared widely; some belongs to a family or community. The person you interview decides. Show a Local Contexts TK Label as an example.
3. **Build questions (15 to 25 min):** Choose five to seven from the protocol; write one original.
4. **Practice in pairs (25 to 40 min):** Partners answer as themselves. Practice the consent script, a follow-up ("Can you tell me more?"), and the thank-you. Switch.
5. **Plan (40 to 46 min):** Who, when, and how you will give back (a copy, a sketch, a thank-you note).
6. **Reflection (46 to 50 min)**

Student handout: Interview Protocol

Read aloud first: "I'm learning about weather, water, and climate where we live. I'd like to ask what you've seen over your life. You can skip any question or stop anytime. I'll only share with my class what you say is okay to share. May I take notes? May I record?" (Record only if they say yes.)

PART	SAMPLE QUESTIONS
Story of Self	Where did you grow up? What was the weather like? What water mattered to you?
Story of Us	What changes have you seen in snow, rivers, fish, smoke, heat, or seasons? What did people do together in hard times?
Story of Now	What do young people need to know? What gives you hope?
Closing	Is there anything you told me that you want kept private? Thank you.

- [ ] I read the consent script and asked before notes or recording.
- [ ] I asked what should stay private.
- [ ] I will share back what I make.

Discussion prompts: Why can an interview be as valuable as a data set? Why might someone decline, and why is that okay?

Differentiation:

- **Grades 6 to 8:** Five given questions plus one original; a family member is a great choice.
- **Grades 9 to 12:** Design the full protocol, add a data pairing plan ("She says the snow comes later; I will check SNOTEL"), and explain in three sentences how the CARE Principles guide what you share.

Formative check: A completed protocol with the consent script and one original follow-up question.

Reflection: (1) What did you learn about how knowledge moves between generations? (2) What shifted in who you think counts as an expert? (3) What one question do you most want to ask your interviewee?

LESSON 2.3

The Elder and the Child Test

- **Grade band and time:** Both; 50 minutes
- **Key lesson:** "The true mark of understanding something is if you can explain to an Elder and a child so that both understand as well." (From the workshop.)
- **Key takeaway:** Clear, respectful, short. If you can say it in 60 seconds with a story and one true fact, it is yours.

Materials: Index cards; sketchbooks; a timer; the checklist below.

Teacher prep: Prepare a 60-second model pitch from your own story. PLACEHOLDER: teacher's own model pitch

Procedure:

1. **Model (0 to 5 min):** Deliver your pitch; ask what students noticed.
2. **Framework (5 to 12 min):** Public narrative (Story of Self, Us, and Now, from organizer and scholar Marshall Ganz). Self: a moment that shows why you care. Us: what we share. Now: what needs to happen, and one ask. Add one accurate fact from Unit 1 or 2.
3. **Draft (12 to 25 min):** Write the pitch on an index card; sketch one image that holds it.
4. **Triads (25 to 40 min):** A speaker, an "Elder listener" (listening as if an Elder you respect were in the room: is it respectful, patient, honest?), and a "child listener" (raising a hand at any unclear word). These are listening roles, not characters. Rotate twice.
5. **Share (40 to 47 min):** Volunteers deliver to the class.
6. **Reflection (47 to 50 min)**

Student checklist: The Elder and the Child Test

- [ ] 60 seconds or less
- [ ] One personal moment (Self) and something we share (Us)
- [ ] One accurate fact, and I know its source
- [ ] One clear action or ask (Now)
- [ ] No unexplained jargon
- [ ] Respectful tone, no doom, ends with hope or action

Discussion prompts: Why is explaining something simply harder than explaining it with big words? Which part was hardest to write: Self, Us, or Now?

Differentiation:

- **Grades 6 to 8:** Sentence starters ("I care about ___ because once ___"; "We all ___"; "Right now we can ___"). A 30-second version, like the Youth Summit's 30-second explanation, is fine.
- **Grades 9 to 12:** Include a data point from the journal or a DDM view with its source; adapt the pitch for a second audience, such as a school board or Tribal youth council.

Formative check: Peers mark the checklist; meeting four of six items is the goal. This is feedback, not a performance grade.

Reflection: (1) What did you learn about explaining climate so others understand? (2) What shifted in how you think about your own voice? (3) What one question will you carry into your interview?

3

UNIT 3

Adaptation, Mitigation, Resilience

Key lesson (from the workshop): These three are not the same. They are different jobs with different tools: policy, groundwork, preparation, data, knowledge, funding, and sometimes emergency.

Key takeaway (from the workshop): The impacts are already here, so we respond quickly and on purpose, without getting lost in fear, the future, and the "black mirror" of media.

LESSON 3.1

The Salmon Journey (Two Periods)

- **Grade band and time:** Both; two periods of 50 minutes
- **Key lesson:** Salmon depend on cold, clean, connected water. Warm water holds less oxygen, and dams, predators, harvest, and hatcheries all shape who comes home.
- **Key takeaway:** In the words of the late Billy Frank Jr., salmon "serve as a measuring stick of our present and future physical and spiritual health" (*Being Frank*, Northwest Indian Fisheries Commission, 03/07/2006). Every choice along the river changes the odds.

PART A: COLD WATER HOLDS MORE LIFE

Materials: Two bottles of sparkling water (one chilled, one room temperature) and two balloons; ice water; warm tap water; thermometers; a dissolved oxygen (DO) kit or probe if available; graph paper.

Teacher prep: Chill one bottle overnight. Download a summer week of Bonneville Dam water temperatures from Columbia River DART (Data Access in Real Time, University of Washington) at cbr.washington.edu/dart, or use a USGS gauge through the DDM.

Procedure:

1. **Hook (0 to 5 min):** In summer 2015, roughly 250,000 adult sockeye died in warm water in the basin; the Nez Perce Tribe's current climate program started after that drought and fish kill. What could make water deadly with no poison in it?
2. **Fizz test (5 to 15 min):** Open both bottles with a balloon over each. The warm one releases more gas: cold water holds dissolved gas better, and oxygen is a gas.
3. **Oxygen and temperature (15 to 30 min):** Measure DO if you have a kit; otherwise graph the table below. Mark 8 mg/L (a common minimum for salmon waters; check your state's standard) and 20 °C.
4. **Real river data (30 to 40 min):** Plot the Bonneville temperatures. How many days crossed 20 °C?
5. **Why is the river warm? (40 to 46 min):** Hot air, low flows after a small snowpack, reservoirs heating in the sun, lost shade. What cools it? Shade, cold tributaries and groundwater (cold-water refuges), more flow.
6. **Reflection (46 to 50 min)**

Student data sheet: Oxygen at Saturation in Fresh Water (Sea Level)

WATER TEMPERATURE	0 °C (32 °F)	5 °C (41 °F)	10 °C (50 °F)	15 °C (59 °F)	20 °C (68 °F)	25 °C (77 °F)
Dissolved oxygen (mg/L)	14.6	12.8	11.3	10.1	9.1	8.3

Saturation is the most oxygen the water can hold. Real rivers often hold less where algae, decay, or slow water use it up.

PART B: THE MIGRATION RELAY

Materials: Eight stations (a chair and station card each); a die per station; salmon cards; two bins labeled "Came Home" and "Fed the People"; chart paper.

Teacher prep: Print station cards. Make salmon cards, half "wild" and half "hatchery," half "Snake River" (farther upstream, more dams) and half "mid-Columbia." Tell students the dice odds are invented; real survival data is at the Fish Passage Center (fpc.org) and CRITFC.

Procedure:

1. **Set up (0 to 8 min):** Each student draws a salmon card and starts at Station 1.
2. **Round 1, cool snowy year (8 to 20 min):** Roll at each station; Snake River fish roll twice at Stations 2 and 7. Lost fish stay and voice that hazard next round.
3. **Round 2, hot low-water year (20 to 30 min):** Hot-year odds. This is what 2015 looked like.
4. **Round 3, restoration year (30 to 40 min):** The class picks three restoration options and reruns the hot year.
5. **Tally and graph (40 to 46 min):** Graph both bins for all rounds.

6. Reflection (46 to 50 min)

Station cards (game odds, not real rates)

STATION	WHAT IT STANDS FOR	COOL YEAR: LOST ON	HOT YEAR: LOST ON	RESTORATION OPTION
1. Home stream	Eggs in the gravel nest (redd); sediment after fire	1	1 to 2	Replant burned slopes: 1 only
2. Downstream dams	Juveniles pass turbines and warm reservoirs with predators	1	1 to 2	Spill water to carry juveniles: 1 only
3. Estuary	Birds and habitat at the river mouth	1	1	Restore wetlands: roll twice, keep the better
4. Ocean	Marine heatwaves, ocean fisheries	1	1 to 2	None: discuss mitigation
5. Bonneville	Sea lions; the fish ladder	1	1	Better ladders: roll twice, keep the better
6. Treaty fishing zone	Tribal Treaty fisheries at usual and accustomed places	On 1 to 2, "Fed the People"	Same	None needed
7. Upstream reservoir	Adults in water above 20 °C	1	1 to 3	Cold-water refuges and shade: 1 only
8. Spawning ground	Home	"Came Home"	Same	Discuss hatchery and wild genetics

Discussion prompts: The "Fed the People" bin is not a loss column. If a Treaty reserves the right to fish (see *United States v. Winans* in the Background), what happens to that right when the fish are gone? Why could the class not change the ocean station, and who can?

Differentiation:

- **Grades 6 to 8:** The teacher offers the Round 3 menu; students make a bar graph.
- **Grades 9 to 12:** Compute percent returning per round; compare the historic estimate of 10 to 16 million salmon and steelhead a year with current counts from CRITFC. Write a claim, evidence, and reasoning paragraph on which station matters most in a hot year.

Formative check: Exit ticket: In a hot, low-water year, what happens to the river's oxygen, which station changed most in the game, and why?

Reflection: (1) What did you learn about what salmon need? (2) Did your sense of what needs to happen, or how fast, change? (3) What one question about salmon in your nearest river will you carry forward?

LESSON 3.2

Adapt, Mitigate, or Build Resilience?

- **Grade band and time:** Both; 50 minutes
- **Key lesson:** Mitigation reduces the causes of climate change. Adaptation adjusts to its impacts. Resilience is the capacity to absorb a shock, recover, and stay who you are. Many real projects do more than one job.
- **Key takeaway:** We must adapt now if we are to be resilient, and mitigate now if we are to limit the impacts on future generations.

Materials: Three hula hoops or yarn loops laid as an overlapping Venn diagram (Mitigation, Adaptation, Resilience); project cards; sticky notes.

Teacher prep: Print the cards without the teacher key.

Procedure:

1. **The flooding room (0 to 7 min):** A faucet is flooding a room. Turning it off is mitigation. Moving things up and mopping is adaptation. Neighbors who help, a plan, and a stocked pantry, so you get through and bounce back, is resilience.
2. **Sort (7 to 25 min):** Teams place each card and write the reason on a sticky note.
3. **Gallery check (25 to 35 min):** Teams visit another Venn and leave one question.
4. **Debrief (35 to 43 min):** Most cards land in overlaps. Which tools did each need: policy, groundwork, preparation, data, knowledge, funding, emergency response? Who decided?
5. **Relational framing (43 to 47 min):** Pick one card: Who? How? When? Why? With whom?
6. **Reflection (47 to 50 min)**

Project cards

Tribal examples come from the *Climate Change and the Tribes* draft (Robinson); verify details with each Tribe before presenting them as current.

#	PROJECT	TEACHER KEY (DEBATE WELCOME)
1	The Yakama Nation moves "problem beavers" from farms and towns to streams, where their dams slow and store water.	Adaptation, Resilience
2	Yakama Nation riverscaping returns logs to the Yakima River so streams reconnect to floodplains and form pools for young salmon.	Adaptation, Resilience
3	The Nez Perce Tribe installs solar with batteries so power stays on in emergencies.	Mitigation, Resilience
4	The Nez Perce Tribe plans to plant nearly 380,000 trees on 2,137 acres (EPA Climate Pollution Reduction Grant, 2024) to store carbon and shade streams.	Mitigation, Adaptation
5	The Nez Perce Tribe plans to install 72 electric vehicle chargers and move its fleet to electric vehicles.	Mitigation
6	The Nez Perce Tribe plans efficiency upgrades, weatherizing, and heat pumps for three Tribal buildings.	Mitigation, Adaptation
7	Nez Perce fisheries work to end creek diversions and keep water cooler.	Adaptation
8	The Confederated Tribes of the Umatilla Indian Reservation (CTUIR) adds a 200-panel solar array at its Tutuilla Food Sovereignty Center, cutting the cost of running its freezers.	Mitigation
9	CTUIR publishes a Climate Adaptation Plan organized around First Foods.	Resilience, Adaptation

#	PROJECT	TEACHER KEY (DEBATE WELCOME)
10	The Confederated Tribes of the Warm Springs Reservation of Oregon and Portland General Electric upgrade a transmission line, with a U.S. Department of Energy grant, to connect reservation renewables.	Mitigation
11	Two Elwha River dams are removed (2011 to 2014) after decades of work led by the Lower Elwha Klallam Tribe.	Adaptation, Resilience
12	The Nez Perce Tribe and others call for removing the four lower Snake River dams; others oppose it over hydropower, barging, and irrigation.	Adaptation, Resilience; how to replace the power cleanly
13	A school builds a rain garden for storm runoff.	Adaptation
14	A city opens cooling centers in a heat wave.	Adaptation (emergency)
15	Old wood stoves are replaced with cleaner stoves certified by the U.S. Environmental Protection Agency (EPA).	Adaptation; debate mitigation
16	The Columbia River Treaty is renegotiated with ecosystem function on the table and Tribes and First Nations participating.	Resilience (governance)
17	A family makes a smoke plan and packs go-bags.	Adaptation, Resilience
18	A school district switches to electric buses.	Mitigation

Discussion prompts: Why do so many Tribal projects land in the overlap? Card 12 has people who disagree; what would a resilient decision process look like?

Differentiation:

- **Grades 6 to 8:** Ten cards (1, 3, 4, 5, 8, 11, 13, 14, 17, 18) with picture icons; definitions posted on the hoops.
- **Grades 9 to 12:** All cards, each labeled with its main tool. Then a real-world skill: many federal programs shifted in 2025 and 2026 from "climate" to "natural hazard and disaster preparedness" language (the Bureau of Indian Affairs, or BIA, Tribal Community Resilience program was formerly Tribal Climate Resilience). Rewrite one card as a two-sentence funding pitch in hazard language without losing its meaning.

Formative check: Place a new example with a reason: "A Tribe plants cottonwoods along a warming creek."

Reflection: (1) How do you now understand adaptation, mitigation, and resilience? (2) Did your sense of what needs to happen, or how fast, change? (3) What one question about responding to climate impacts in your community will you carry forward?

LESSON 3.3

Virtual Field Study on the Dynamic Drought Module

- **Grade band and time:** Both; 50 minutes
- **Key lesson:** Data is a way of knowing too. Data literacy is sovereignty: knowing what data exists, what it can and cannot tell you, and who controls it.
- **Key takeaway:** Understand data, know what data is useful, and know how data leads to action.

Materials: A device per pair; the data sheet and framing worksheet below.

About the tool: The ATNI Dynamic Drought Module (DDM) is public and free, with no login: https://atniclimate.github.io/dynamic-drought-module/ . It shows live drought, wildfire, heat, streamflow (USGS gauges), snowpack (SNOTEL), NWS alerts, the Climate Prediction Center (CPC) seasonal drought outlook, and BIA reservation boundaries; clicking a boundary opens a place-based impact briefing. Every view is a shareable link. It supports decisions; it never replaces Tribal decision-making.

Teacher prep: Open the Columbia and Snake preset: https://atniclimate.github.io/dynamic-drought-module/?region=columbia_snake_basin . Turn on layers; the link updates to record them (the `layers` parameter). Post that link so everyone starts from the same view; add `&embed=true` for a clean view in a slide or learning management system. Pick three USGS gauges and two SNOTEL sites near your students, and confirm the site loads on your school network.

Procedure:

1. **Frame (0 to 5 min):** We cannot all go to the river today, so it comes to us through gauges, snow sensors, and satellites. Who put these sensors here? Who decides what gets measured?
2. **Tour (5 to 12 min):** Show each layer; click a reservation boundary and read part of the briefing. Explain the empty Tribal layers: ATNI's Tiered Sovereign Data Framework treats public data (the "Rain" tier) as the base and keeps culturally sensitive data (the "Canoe" tier) under each Nation's control.
3. **Stations (12 to 35 min):** Pairs complete the data sheet.
4. **Framing worksheet (35 to 43 min):** Pairs work through one issue they found.
5. **Share a view (43 to 47 min):** Each pair posts its best view's link with a one-sentence caption.
6. **Check-in and reflection (47 to 50 min):** One word for how you feel after this data, then one action.

Student data sheet: Columbia Basin Virtual Field Study

STATION	WHAT TO FIND	WHAT I FOUND	WHAT IT MEANS
1. Drought	Drought level where you live		
2. Streamflow	One gauge: flow now compared with normal		
3. Snowpack	One SNOTEL site compared with median (zero in late summer: why?)		
4. Heat and fire	Heat alerts or fire perimeters in the basin		
5. Outlook	Seasonal drought outlook for your area		
6. Briefing	Two impacts from one reservation's briefing		

Student handout: Framing Worksheet (adapted from Tribal College and University student research practice)

STEP	QUESTION	MY NOTES
What's Known	What does the data show?	
What's Under Threat	Health, land, economy, community, ways of life?	
Projected Impacts	What could happen next season or in coming decades?	

STEP	QUESTION	MY NOTES
Methods	How would you find out more? What would you measure? Whom would you ask?	
Actions	Adapt, mitigate, or build resilience?	

Discussion prompts: What does this map show well, and what is missing (local observations, Indigenous Knowledges, water quality, groundwater)? Why might a Nation choose not to put its data on a public map?

Differentiation:

- **Grades 6 to 8:** Stations 1 to 3 with a teacher-chosen gauge and site; framing worksheet as a class.
- **Grades 9 to 12:** All six stations, plus a 150-word data brief for a Tribal natural resources department or school board citing the shared link.

Teacher note: Data changes daily. The DDM is not an alert system; use weather.gov for warnings.

Formative check: A framing worksheet that ties at least one data point to one action.

Reflection: (1) What did the data teach you about the basin right now? (2) What shifted in how you think about who collects and controls data? (3) What one question would you ask a sensor, or a knowledge holder?

4

UNIT 4

Relationships Matter

Key lesson (from the workshop): It's going to take all of us. We can do more together than any of us can alone, and that includes our relationship with our non-human relatives.

Key takeaway (from the workshop): We are not apart from nature; we are a part of nature. Go outside, pay attention, be a good relative, be a good ancestor. Our world is changing rapidly, but we have all that we need, and we are sovereign, and we decide what matters. We are not born into this world; we are born out of it.

LESSON 4.1

Who Is Connected to Whom?

- **Grade band and time:** Both; 50 minutes
- **Key lesson:** Every Being and every group in the watershed is tied to others. A stress on one travels through the web, and so does care.
- **Key takeaway:** Relatives, not resources.

Materials: A sturdy ball of yarn; role cards; the basin map; sketchbooks; approved interview shares from Lesson 2.2.

Teacher prep: Print role cards; clear space for a standing circle; confirm each interview share was approved.

Procedure:

1. **Roles (0 to 5 min):** Students stand in a circle wearing role cards.
2. **Build the web (5 to 20 min):** Snowpack starts ("I am snowpack. I feed the river in summer"), holds the end, and tosses the yarn to River. Each holder names a real connection and tosses on, keeping their strand, until everyone holds two.
3. **Stress events (20 to 28 min):** The named role tugs; anyone who feels it says how. Heat wave warms the river past 20 °C (River). Snow drought (Snowpack). Wildfire above a spawning stream (Wildfire crew). A dam without fish passage (Dam operator). A decision made without consulting the Tribe (Tribal Nation government): a social strand, and it still moves the whole web.
4. **Care events (28 to 36 min):** Beavers return; shade trees are planted; Treaty rights are upheld and Tribes co-manage from the start. Add a new strand for each.
5. **Interview share-in (36 to 44 min):** Students share one approved sentence from their interview and add a strand from "Elder or knowledge holder."
6. **Record (44 to 47 min):** Photograph or sketch the web.
7. **Reflection (47 to 50 min)**

Role cards: Snowpack; Glacier; Rain; River; Groundwater; Chinook salmon; Pacific lamprey; White sturgeon; Southern Resident orca; Beaver; Cottonwood; Huckleberry; Bald eagle; Sea lion; Mayflies and caddisflies; Soil; Tribal fisher; Tribal Nation government; CRITFC scientist; Farmer and irrigator; Dam operator; Electricity customer; City water plant; Wildfire crew; School; Student; Elder or knowledge holder; First Nation in the Canadian upper Columbia; State agency; Federal agency. Tribal roles are positions in a system, not identities to act out.

Discussion prompts: Which roles got tugged by almost every stress, and what does that tell you? Why did the consultation strand move the whole web?

Differentiation:

- **Grades 6 to 8:** Print a sample connection sentence on the back of each card.
- **Grades 9 to 12:** Label each strand ecological, economic, legal (Treaty), or cultural; redraw the web as a network diagram and find the most connected roles.

Formative check: A sketch of the web with five labeled strands and one sentence tracing how a stress traveled.

Reflection: (1) What did you learn about how the watershed is connected? (2) What shifted in how you see your own place in it? (3) What one relationship do you want to learn more about?

LESSON 4.2

Two Rivers: The Elwha and the Columbia

- **Grade band and time:** Both; 50 minutes
- **Key lesson:** On the Elwha, the Lower Elwha Klallam Tribe led a decades-long push that ended in the largest dam removal in U.S. history at the time. The Columbia is far larger and more complex, and its choices differ.
- **Key takeaway:** Change that once seemed impossible happened. Stories of repair are evidence too.

Materials: PBS *Nature*, "The Beautiful Undammed" (*Wild Hope*): https://www.pbs.org/wnet/nature/the-beautiful-undammed-wild-hope/29052/ ; National Park Service, Elwha River Restoration: https://www.nps.gov/olym/learn/nature/elwha-ecosystem-restoration.htm ; projector; sketchbooks; comparison chart.

Teacher prep: Preview the film and check its running time; plan up to 25 minutes or select segments. Pick two tour stops. Fact box: Elwha Dam was completed in 1913 and Glines Canyon Dam in 1927; Congress passed the Elwha River Ecosystem and Fisheries Restoration Act in 1992; both dams came out from 2011 to 2014. For Lower Elwha Klallam history, use the Tribe's own published materials.

Procedure:

1. **Frame (0 to 5 min):** Fold a page into three panels: Before, During, After. As at the Youth Summit, the assignment is to take notes in sketches.
2. **Watch (5 to 30 min):** Film or segments, sketch-noting throughout.
3. **Virtual tour (30 to 36 min):** Two Washington Sea Grant stops: What changed after the dams came out?
4. **Compare (36 to 45 min):** Pairs fill the chart.
5. **Share and reflect (45 to 50 min)**

Student handout: Two Rivers Comparison

QUESTION	ELWHA RIVER	COLUMBIA RIVER
Size of river and watershed		
Number and purpose of dams		
Who led the push for change?		
Legal tools used		
How long did it take?		
What came back?		Not yet known
What could transfer, and where does the comparison break down?		

Discussion prompts: What did it take on the Elwha: time, law, science, relationships, persistence? What makes the Columbia different (scale, hydropower, navigation, irrigation, an international treaty, many governments)?

Differentiation:

- **Grades 6 to 8:** Three-panel timeline and a simple Venn diagram.
- **Grades 9 to 12:** Research how scientists tracked sediment after removal; write an argument on what the Columbia can learn from the Elwha and where the comparison fails.

Formative check: A completed timeline and one sentence: "One lesson from the Elwha the Columbia could use is ___ because ___."

Reflection: (1) What did you learn from the Elwha story? (2) What shifted in what you believe is possible for a river? (3) What one question about the Columbia's future will you carry forward?

LESSON 4.3 DRAFT

Good Energy, Wrong Place? A Consultation on Goldendale and Pushpum (Two Periods)

Draft, under review. Not ready to teach. This lesson asks students to role-play Tribal governments at a consultation table. ATNI Climate is revising it so that no student speaks as a Nation, council, Elder, or knowledge holder. It is printed here for facilitator review only.

- **Grade band and time:** 9 to 12 (one-period 6 to 8 version below); two periods of 50 minutes
- **Key lesson:** Clean energy is needed, and Treaty rights and sacred places are real. How a decision is made, and who is at the table from the start, matters as much as what gets built.
- **Key takeaway:** Good process is climate action too. A Tribal leader quoted in the *Climate Change and the Tribes* draft (Robinson) said the Tribes are tired of being left out of dinner only to be invited to dessert.

This is a live issue. The Goldendale pumped storage project is real and ongoing. The Federal Energy Regulatory Commission (FERC) issued a 40-year license on 01/22/2026; rehearing was denied in 03/2026; the Yakama Nation and Columbia Riverkeeper petitioned the Ninth Circuit. On 09/21/2026 the Washington Court of Appeals sent part of the state water quality certification back to the Department of Ecology to add leak monitoring. Verify status before teaching, and check any quotation from the FERC proceeding against the FERC order or a Yakama Nation statement.

Be open about sources. These lessons come from ATNI Climate, and the draft reports that ATNI and the National Congress of American Indians (NCAI) passed resolutions opposing the destruction of Pushpum. Tell students plainly; naming it is part of Source Check.

Case summary (from the draft; verify all details):

- **The proposal:** Rye Development proposes pumped storage hydropower near John Day Dam in Klickitat County, Washington, on and around a former aluminum smelter site. Pumped storage uses surplus electricity to pump water uphill, then runs it through turbines when power is needed, which can help a grid that relies on wind and solar.
- **Supporters' case:** The draft reports claims that it could serve up to half a million homes without greenhouse gas emissions and reuse an industrial site, and reports a link to a proposed data center.
- **Findings:** The draft cites a Washington Department of Ecology review finding "significant and unavoidable adverse impacts" to Yakama historic sites and culturally significant plants.
- **The place:** Within Yakama Nation ceded lands and, per the draft, used by the Yakama Nation, CTUIR, Warm Springs, and the Nez Perce Tribe. The draft gives the Yakama name as Pushpum, "Mother of all roots."
- **Opposition:** The Yakama Nation opposes the project; the draft reports opposition from other Tribes and conservation groups.

Materials: Case summary; role packets; decision frame; name tents.

Teacher prep: If Yakama citizens or citizens of other affected Nations are in your class, talk with them and their families first; never ask a Native student to represent a Nation. Gather public sources for each role: the Yakama Nation's statements PLACEHOLDER: current Yakama Nation public statements on Pushpum, the developer's materials, Ecology documents, and FERC records. **Quote, don't perform:** students representing a Tribal Nation paraphrase that Nation's public statements and never improvise cultural or spiritual content about Pushpum.

Roles

ROLE	THEIR PUBLIC CASE
Yakama Nation Tribal Council (a sovereign government)	Treaty of 1855; ceded lands; a sacred site; consultation
Other Treaty Tribes with historic use (CTUIR, Warm Springs, Nez Perce)	Their own public positions; do not assume
The developer	Clean energy storage, grid reliability, jobs, site reuse

ROLE	THEIR PUBLIC CASE
Grid and clean energy planners	Washington's Clean Energy Transformation Act requires 100 percent clean electricity by 2045; storage helps integrate wind and solar
Washington Department of Ecology	Its impact findings; what mitigation can and cannot do
FERC	Licensing; consultation under Section 106 of the National Historic Preservation Act; the federal trust responsibility
County residents and workers	Jobs, tax base, local concerns; a range of views
Conservation and climate groups	Some oppose this project; some support storage generally

Period 1:

1. **Frame (0 to 10 min):** Live issue note and source disclosure. What do "clean energy" and "consultation" mean?
2. **Read the case (10 to 20 min):** Mark what is reported and what is verified.
3. **Role research (20 to 45 min):** Draft a two-minute statement from public sources.
4. **Check-in (45 to 50 min):** One word for how you feel.

Period 2:

1. **Set the room (0 to 5 min):** A government-to-government table (FERC and the Tribal Nations) and a separate public comment area: consulting a sovereign is not the same as hearing the public.
2. **Consultation (5 to 25 min):** FERC opens and listens. The Yakama Nation speaks first, then other Tribal Nations, then public comment, two minutes each.
3. **Out of role (25 to 35 min):** Talking circle: What did you hear? What surprised you?
4. **Recommendation (35 to 45 min):** Each student writes their own, using the decision frame.
5. **Reflection and care check (45 to 50 min)**

Student handout: Decision Frame

1. What is the need, and for whom?
2. What rights are involved? (Treaty rights reserved by the Tribes; sacred sites; federal consultation duties; free, prior, and informed consent under Article 32 of the United Nations Declaration on the Rights of Indigenous Peoples, which the U.S. endorsed in 2010.)
3. What harms are called "unavoidable," and who carries them?
4. What alternatives exist? (Other sites, other storage technologies, other ways to meet demand.)
5. Who should decide, and when should they have been at the table?
6. My recommendation, reasons, and sources.

Discussion prompts: Can clean energy goals and Treaty rights both be honored, and what would that take? The draft sums it up: clean energy, good; on sacred ground, bad; without Tribal consultation, bad. Do you agree, and what evidence did you weigh?

Differentiation:

- **Grades 6 to 8 (one period):** Five roles (Yakama Nation, developer, grid planners, state agency, youth council), a statement template, and the decision frame done as a class.
- **Grades 9 to 12:** Both periods; a 300-word recommendation citing three sources, with Source Check applied to at least one "clean energy" claim.

Formative check: A recommendation that accurately describes at least two other perspectives and addresses process, not only outcome.

Reflection: (1) What did you learn about how energy decisions get made? (2) What shifted in how you think about "clean" energy? (3) What one question about this case will you keep following?

Closing Lesson

CLOSING

From Learning to Leading (Two Periods)

- **Grade band and time:** Both; two periods of 50 minutes
- **Key lesson:** "We will always be better than I ever will be." No one carries this work alone, and every path (science, fire, law, teaching, art, trades, health) is part of it.
- **Key takeaway:** Be a good relative, be a good ancestor. Our world is changing rapidly, but we have all that we need.

Materials: Chart paper career stations; sticky notes; commitment cards; sketchbooks with the Opening half circle; the talking piece.

Teacher prep: Set up stations. If possible, invite a professional through the local Tribe or a nearby Tribal College, and compensate them. PLACEHOLDER: guest professionals PLACEHOLDER: regional TCU, college, and internship programs Plan the closing circle with the Tribe's guidance. PLACEHOLDER: closing per the local Tribe's guidance; no traditional song or blessing unless offered by an invited knowledge holder

PERIOD 1: CAREER PATHWAYS GALLERY

1. **Frame (0 to 5 min):** Every lesson in this set is someone's job.
2. **Gallery walk (5 to 30 min):** Pairs leave a sticky note at each station: "A question I'd ask this person."
3. **Guest or video (30 to 40 min):** Questions for a guest, or a short video of a professional at work.
4. **Pathway sketch (40 to 47 min):** Three steps from today toward one role.
5. **Reflection (47 to 50 min)**

Career stations (expanded from the Youth Summit's Module 6)

STATION	WHAT THEY DO
Environmental scientist	Water quality monitoring, restoration planning
Wildland firefighter or emergency manager	Fire response, preparedness
Tribal natural resources professional	Fisheries, wildlife, forestry, water, alongside knowledge holders
Hydrologist or fisheries biologist	Streamflow, temperature, fish counts
Geographic information systems (GIS) analyst	Maps, sensors, data sovereignty
Energy technician or lineworker	Solar, storage, transmission, weatherization
Policy advocate or attorney	Treaty rights, consultation, climate law
Educator	Teaching and science communication
Artist, writer, or filmmaker	Telling the story so people act
Community health worker	Heat and smoke preparedness

PERIOD 2: COMMITMENTS AND THE SECOND HALF OF THE CIRCLE

1. **Commitments (0 to 10 min):** Fill in the commitment card.
2. **Second half of the circle (10 to 30 min):** Return to the Opening half circle and draw the bottom half, from today through the rest of your life: hopes and plans around climate, water, and your homeplace. Plans will change; that is fine.

3. **Closing circle (30 to 45 min):** With the talking piece, each student shares one thing from their circle and one commitment. Close with gratitude for the lands, the waters, and each other.
4. **Final reflection (45 to 50 min)**

Student handout: Commitment Card

PROMPT	MY COMMITMENT
One thing I will do differently	
How and with whom I will share what I learned	
What I want to keep learning	
One next step toward a path or career	
How our group will support each other, and one thing we will do together	

Discussion prompts: Who will you bring into this work with you? What does it mean to be a good ancestor?

Differentiation:

- **Grades 6 to 8:** Home- and school-scale commitments, chosen from a menu.
- **Grades 9 to 12:** Add one civic action (public comment, a letter to a council, a watershed group or Tribal youth council) and one skill to build; option to present the plan, as at the Youth Summit.

Formative check: A completed commitment card and a finished circle.

Final reflection (whole set): (1) What did you learn across these lessons? (2) What idea shifted most for you? (3) What one question will you carry forward?

Columbia River Background for Teachers

Footing for student questions and for catching errors in other materials.

The River and Its Basin

The Columbia rises at Columbia Lake in British Columbia and runs about 1,240 miles to the Pacific, forming much of the Washington and Oregon border. Its basin covers roughly 260,000 square miles in seven U.S. states and British Columbia; the Snake River is its largest tributary. Much of its water begins as mountain snow. East of the Cascades, in the rain shadow, rivers depend on snowmelt, glaciers, and groundwater to get through dry summers.

Peoples of the River and Reserved Rights

Many Peoples have lived with the Columbia and its tributaries since time immemorial. These lessons focus on the four Treaty Tribes of the Columbia River Inter-Tribal Fish Commission (CRITFC): the Confederated Tribes of the Umatilla Indian Reservation (CTUIR), the Confederated Tribes of the Warm Springs Reservation of Oregon, the Confederated Tribes and Bands of the Yakama Nation, and the Nez Perce Tribe. Many other Nations hold deep relationships with the basin, among them the Confederated Tribes of the Colville Reservation, the Spokane Tribe, the Coeur d'Alene Tribe, the Kalispel Tribe, the Kootenai Tribe of Idaho, the Confederated Salish and Kootenai Tribes, the Shoshone-Bannock Tribes, the Cowlitz Indian Tribe, the Chinook Indian Nation, and First Nations in Canada including the Syilx Okanagan, Ktunaxa, and Secwepemc. Learn whose homelands include your school.

In 1855 the United States negotiated Treaties with the Yakama; the Nez Perce; the Walla Walla, Cayuse, and Umatilla; and, in the Treaty with the Tribes of Middle Oregon, the Tribes now of Warm Springs. The Tribes ceded millions of acres and reserved (kept) rights, including the right of taking fish at all usual and accustomed places and rights to hunt and gather. In *United States v. Winans* (1905), the U.S. Supreme Court called the Treaty "not a grant of rights to the Indians, but a grant of right from them, a reservation of those not granted." This is the reserved-rights doctrine.

Federal courts have repeatedly upheld these rights. In *United States v. Oregon*, Judge Robert Belloni ruled in 1969 that the four Columbia River Treaty Tribes have a right to a fair share of the fish runs, and in 1974 he applied the Boldt 50/50 principle to define that share; the case continues through court-supervised management agreements among the Tribes, states, and federal government. In *United States v. Washington* (1974), Judge George Boldt held that Treaty Tribes are entitled to up to 50 percent of the harvestable fish. The four Treaty Tribes formed CRITFC in 1977. Tribes are co-managers of these fisheries.

Salmon and a Changed River

An estimated 10 to 16 million salmon and steelhead once returned to the Columbia each year. Canneries opened in 1866 and drove heavy overfishing. In the twentieth century the Columbia became one of the most heavily dammed river systems in the world: 14 dams on the main stem (11 in the U.S., 3 in Canada), four on the lower Snake (Ice Harbor, Lower Monumental, Little Goose, and Lower Granite), and hundreds across the basin. Grand Coulee (completed 1942) and Chief Joseph dams have no fish passage and cut salmon off from the upper Columbia; Grand Coulee's reservoir flooded the great fishery at Kettle Falls. Upper Columbia Tribes and First Nations are leading work to return salmon above these dams (verify status). The Hells Canyon dams block the upper Snake. On 03/10/1957, The Dalles Dam flooded Celilo Falls, a fishing and trading place used for thousands of years.

Hatcheries now produce a large share of returning fish (the draft says most; verify with CRITFC). They support harvest and raise real questions about wild genetics. The Hanford Site produced plutonium from World War II through the Cold War; about 56 million gallons of radioactive and chemical waste remain in 177 underground tanks, and the Yakama Nation, CTUIR, and the Nez Perce Tribe take part in cleanup oversight.

Climate Change on the Columbia

- **Global context:** The recent decade averaged about 1.1 to 1.3 °C (2.0 to 2.4 °F) above the 1850 to 1900 average. The World Meteorological Organization reported single years at about 1.45 °C (2023) and 1.55 °C (2024).
- **Snow and timing:** The Fifth National Climate Assessment (2023), Chapter 27 (Northwest), documents warming, declining spring snowpack, and shifting streamflow timing. In the 2015 snow drought, near-normal precipitation in many places fell as rain, leaving rivers low and warm by summer.
- **Water temperature and salmon:** Adult migration is impaired above roughly 20 to 21 °C (68 to 70 °F), and warm water holds less oxygen. In summer 2015, roughly 250,000 adult sockeye died in warm water in the basin. Cold-water refuges at tributary mouths matter more every year.
- **Heat:** The late June 2021 heat dome set all-time records across the Northwest, including 116 °F in Portland.
- **Fire and smoke:** The 2017 Eagle Creek Fire burned about 50,000 acres in the Columbia River Gorge; the Burdoin Fire started across the river on 07/18/2025, east of Bingen, between Bingen and Lyle, in Klickitat County. After fires, erosion can bury salmon nests in sediment.
- **Glaciers and ocean:** Cascade glaciers are shrinking; on the west slope, the Hinman Glacier was gone by 2022. Marine heatwaves and ocean acidification affect young salmon's food web.
- **First Foods:** CTUIR organizes its climate work around First Foods (water, salmon, deer, cous, and huckleberry). Its Climate Adaptation Plan summary is public: https://ctuir.org/media/oeklm2el/cap-executive-summary-final-std-pg-no.pdf

Governance and Policy Now (Verify at Time of Teaching)

- **Columbia River Treaty:** The 1964 U.S. and Canada treaty governs flood control and hydropower. Tribes and First Nations were excluded from its negotiation. U.S. Tribes have participated in the Treaty review since 2012, and the Syilx Okanagan, Ktunaxa, and Secwepemc Nations observed with the Canadian delegation. An Agreement in Principle on modernization was announced 07/11/2024; the U.S. paused negotiations in March 2025 amid trade tensions (https://www.opb.org/article/2025/03/14/us-pauses-columbia-river-water-sharing-agreement-with-canada/).
- **Resilient Columbia Basin Agreement:** A December 2023 agreement among the United States, the four Treaty Tribes, Oregon, and Washington. The federal government withdrew on 06/12/2025, by executive action, without Tribal consultation (https://critfc.org/2025/06/12/21153/).
- **Federal funding (volatile):** The Federal Emergency Management Agency (FEMA) ended its Building Resilient Infrastructure and Communities (BRIC) program in April 2025; a court ordered it restarted, and FEMA in 2026 describes reconstituting it. The BIA Tribal Community Resilience (TCR) Annual Awards Program lists FY24 as its latest cycle (https://www.bia.gov/service/tcr-annual-awards-program). EPA's Environmental and Climate Justice Block Grants were terminated in 2025; Tribes are among plaintiffs in *Appalachian Voices et al. v. EPA*, on appeal as of early 2026 (https://earthjustice.org/press/2026/nonprofits-tribes-and-local-governments-appeal-to-restore-epa-grant-program).

Standards Alignment

Verify codes against your state framework. Codes are Next Generation Science Standards* performance expectations unless noted.

STANDARD	FOCUS	LESSONS
MS-ESS2-4	Water cycling driven by sunlight and gravity	1.1, 1.3
MS-ESS2-5	Air masses and changes in weather	1.1, 1.2
MS-ESS3-3	Monitoring and reducing human impact	3.2, 4.1, 4.2
MS-ESS3-5	Factors behind rising global temperatures	2.1, 3.3
MS-PS1-4	Thermal energy, particle motion, and changes of state	1.1, 3.1 A
MS-LS2-4	Changes to ecosystem components affect populations	3.1, 4.1
HS-ESS2-4	Changes in energy flow change climate	2.1, Background
HS-ESS2-5	Properties of water and surface processes	1.3, 3.1 A
HS-ESS3-1	Resources, hazards, and climate shape human activity	3.3, 4.3
HS-ESS3-4	Evaluating solutions that reduce human impacts	3.2, 4.3
HS-ESS3-5	Analyzing geoscience data and climate forecasts	1.3, 3.3
HS-LS2-7	Solutions that reduce impacts on biodiversity	3.1 B (Round 3), 4.2
Washington: *John McCoy (lulilaš) Since Time Immemorial* (Office of Superintendent of Public Instruction, OSPI; RCW 28A.320.170, as amended 2024)	Tribal sovereignty curriculum (verify unit alignment in OSPI materials)	Opening, 3.1, 4.1, 4.3, Background
Oregon: Senate Bill 13, Tribal History/Shared History (Oregon Department of Education, with Oregon's nine federally recognized Tribes)	Essential Understandings of Native Americans in Oregon: 1 Since Time Immemorial, 2 Sovereignty, 6 Lifeways, and 8 Treaties with the United States (nine in all, ODE 2020)	Opening, 3.1, 4.3, Background
Idaho and Montana	PLACEHOLDER: state or Tribal education alignment	
Common Core English Language Arts, Speaking and Listening (SL.1, SL.4)	Collaborative discussion; presenting claims	2.3, 4.3, Closing

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Resources

Tools

- ATNI Dynamic Drought Module: https://atniclimate.github.io/dynamic-drought-module/ (preset ? `region=columbia_snake_basin`)
- Other ATNI Climate tools (Plan Assessor, TCR Policy Scanner, Policy Sentinel, ATNI-GeoBase): PLACEHOLDER: current access contact for ATNI Climate tools; Policy Sentinel code: github.com/atniclimate/policy-sentinel
- earth.nullschool.net; National Weather Service: https://www.weather.gov/ ; AirNow: https://www.airnow.gov/
- CoCoRaHS: https://www.cocorahs.org/ ; U.S. Drought Monitor: https://droughtmonitor.unl.edu/
- Columbia River DART: https://www.cbr.washington.edu/dart ; Fish Passage Center: https://www.fpc.org/
- The Climate Toolbox and Tribal Climate Tool (Northwest Climate Adaptation Science Center, or NW CASC; University of Washington Climate Impacts Group; University of Idaho): https://climatetoolbox.org/

Tribal and intertribal sources

- CRITFC: https://critfc.org/ ; Northwest Indian Fisheries Commission: https://nwifc.org/
- CTUIR Climate Adaptation Plan summary: https://ctuir.org/media/oeklm2el/cap-executive-summary-final-std-pg-no.pdf
- Nez Perce Tribe: https://nezperce.org ; ATNI: https://atnitribes.org
- PLACEHOLDER: local Tribe education and cultural department contacts

Climate and Tribal resilience

- Tribal Climate Change Guide, University of Oregon: https://tribalclimateguide.uoregon.edu/
- Tribal Resilience Actions database: https://tribalresilienceactions.org/plans/
- Institute for Tribal Environmental Professionals (ITEP), Tribal Wellbeing for Seven Generations Program, Northern Arizona University: https://itep.nau.edu/tw7g/
- NW CASC: https://nwcasc.uw.edu/
- Fifth National Climate Assessment (2023), Chapter 16, "Tribes and Indigenous Peoples," and Chapter 27, "Northwest": doi:10.7930/NCA5.2023.CH16 and doi:10.7930/NCA5.2023.CH27. The federal site went offline in mid-2025; an online copy hosted by the nonprofit Climate.us is at https://nca5.climate.us/

Elwha River

- PBS *Nature*, "The Beautiful Undammed" (*Wild Hope*): https://www.pbs.org/wnet/nature/the-beautiful-undammed-wild-hope/29052/
- National Park Service, Elwha River Restoration: https://www.nps.gov/olym/learn/nature/elwha-ecosystem-restoration.htm

Working with Tribes and Indigenous data

- Local Contexts (TK and BC Labels): https://localcontexts.org/
- CARE Principles, Global Indigenous Data Alliance: https://www.gida-global.org/careprinciples